

Nutrition & Healthy Eating Knowledge Base

Beginner RAG knowledge document for the AI Health & Nutrition Assistant

1. Introduction to Nutrition

Nutrition is the process through which the body obtains and uses nutrients from food. A balanced eating pattern generally includes a variety of vegetables, fruits, whole grains, legumes, protein-rich foods, and appropriate sources of healthy fats. Nutritional needs vary with age, body size, activity, goals, health status, and individual dietary requirements.

2. Macronutrients

Nutrient	Main role	Common sources
Protein	Growth, maintenance and repair of body tissues.	Dal, beans, chickpeas, soy, tofu, paneer, eggs, fish, meat, dairy
Carbohydrate	Important source of energy.	Rice, roti, oats, potatoes, fruits, vegetables, whole grains
Fat	Provides energy and helps absorb some vitamins.	Nuts, seeds, oils, avocado, dairy, fish

3. Micronutrients

Vitamins and minerals are needed in smaller amounts but are important for normal body functions. Examples include iron, calcium, vitamin D, vitamin B12, folate, vitamin C, potassium and magnesium. A varied diet can help provide many micronutrients.

4. Fiber

Dietary fiber is found mainly in plant foods. Good sources include vegetables, fruits, whole grains, beans, lentils, chickpeas, nuts and seeds. Increase fiber gradually and maintain adequate fluid intake.

5. Hydration

Water is essential for normal body functions. Fluid needs vary with body size, climate, activity, diet and other factors. Drink regularly throughout the day. Individual medical conditions may require different fluid guidance.

6. Healthy Meal Planning

A practical meal can combine several food groups: a whole-grain source, a protein-rich food, vegetables or fruit, and a suitable fat source. Portion sizes should be adjusted to individual energy needs and goals.

7. Protein-Rich Vegetarian Foods

Useful options include lentils (dal), chickpeas (chana), rajma, soy, tofu, paneer, curd/yogurt, nuts and seeds. Their nutritional values vary by product and preparation.

8. Common Indian Meal Options

Examples include poha, upma, idli, dosa, roti, rice, dal, rajma, chole, vegetables, curd, paneer, tofu, sprouts, fruits, nuts and seeds. Nutritional value depends on ingredients, preparation method and portion size.

9. General Weight-Management Principles

Choose mostly nutrient-dense foods; consider portion sizes and overall energy intake; include protein-rich foods regularly; eat vegetables, fruits and fiber-rich foods; limit excessive added sugars and excess sodium; and combine nutrition with physical activity, adequate sleep and other healthy habits.

10. Food Allergies and Dietary Restrictions

A food identified as an allergen should not be recommended by the application. Users with allergies should verify ingredients and cross-contact risks. Vegetarian, vegan, cultural and other dietary preferences should be respected.

11. RAG Guidance

This is a demonstration knowledge base for a Retrieval-Augmented Generation application. A production health application should use current, authoritative sources, retain source metadata and publication dates, and review and update the knowledge base regularly.

Safety Disclaimer

This document provides general nutrition education. It is not a medical diagnostic or treatment resource. People with medical conditions, pregnancy, food allergies, eating disorders, or other special requirements should seek advice from an appropriately qualified healthcare professional.